

Technical Paper:

A Method of Designing and Fabricating McKibben Muscles Driven by 7 MPa Hydraulics

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Research has recently been increasing on light weight and high-power robot hands that use artificial muscles. By applying ultra high strength PBO fiber sleeves to McKibben artificial muscles, new hydraulic artificial muscles have been developed in our laboratory. In this research, to apply this technology to a high-power robot easily, we have developed new, thin, hydraulic artificial muscles. While the hydraulic artificial muscles reported in our previous paper were driven by a maximum water pressure of 4 MPa, the newly developed thin muscles are driven by water with a maximum pressure of 7 MPa, resulting in very high force capability. This paper details the materials and structure of the new artificial muscles and reports the results of experiments on them. The muscles developed in this work are based on a sleeve and rubber tube design. The movements of the muscles depend on the angle of the knit of sleeve: an angle of less than 54.5 deg produces contraction while an angle of more than 54.5 deg produces extension. Based on this idea, we optimize, using FEM analysis, the angle of knit of the sleeve of each muscle. As a result, a high powered artificial muscle 21 mm in diameter which generates 8 kN of contraction force has been successfully developed.

Keywords: McKibben artificial muscle, rubber actuator, hydraulic actuator, power robot

1. Introduction

Compact yet high powered robots are needed for rescue, forestry and welfare operations. Recently, many types of actuators have been developed [1, 2], but hydraulic cylinders in particular have been applied to several types of robots and tools [3, 4]. Large robots, such as construction machines, are generally driven by hydraulic cylinders that generate high levels of force [5].

We previously developed large, hydraulic, artificial muscles [3, 4], as can be seen in **Fig. 1**. In our research, we have developed thinner and more powerful hy-



(a) Slim muscle ($\phi = 24$ mm)



(b) Large muscle ($\phi = 40$ mm)

Fig. 1. Hydraulic artificial muscles.

draulic muscles for power robots through the following four technical key points: (1) applying recently-developed super fibers, PBO (p-phenylene-2, 6-benzobisoxazole) fibers, [6] for the sleeve, (2) analyzing and designing suitable patterns for the weaving sleeves for high-force muscles, (3) developing three-layer tubes for pressure-resistance improvement, and (4) developing new shield plug structures for the ends of the muscles. Based on these key points, we have developed two types of artificial muscles and carried out the performance evaluations of them.

2. Working Principle and Characteristics of McKibben Artificial Muscles

2.1. Structure of the Artificial Muscle

The working principle and structure of McKibben artificial muscle have been adapted to the new artificial muscle. The McKibben muscle was originally invented and developed in the 1950s [7] and it has been modified and commercialized [8, 9]. This muscle consists of a rubber tube, a sleeve weaved in spiral and shield plugs at the both ends of the rubber tube. **Fig. 2** shows the general structure of the artificial muscle. Expansion of the rubber tube caused by air pressure results in contract on force in the axial direction. The working characteristics of this muscle strongly depend on the pressure applied and the angle of the spiral of the sleeve fibers [7].

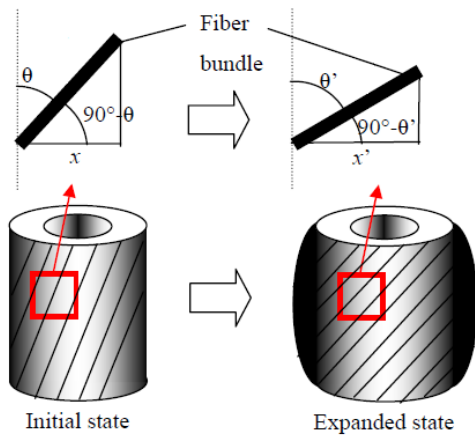


Fig. 2. Model of the change in the angle of the spiral caused by the expansion of the artificial muscle.

2.2. Theoretical Force of Contraction

McKibben muscles generate contraction forces generally expressed by the following Eq. (1), but the equation can also be applied to hydraulic artificial muscles. Additionally, the equation is effective not only for contraction forces but also for extension forces [8].

$$F = \frac{\pi}{4} D_0^2 P \left(\frac{1}{\sin \theta_0} \right)^2 \left\{ 3(1 - \varepsilon)^2 \cos^2 \theta_0 - 1 \right\} \quad (1)$$

D_0 , P , θ_0 , and ε denote the diameter of the artificial muscle, the applied pressure, the initial angle of the knit of the sleeve, and the contraction rates, respectively. This equation indicates that the contraction rate and the generated forces vary with the initial angle of knit. Given that forces generated are proportional to pressure applied, the developed artificial muscles, using water instead of air as working fluid to apply high pressure, can generate large amounts of force [10, 11].

3. Analysis and Design of Three-Layer Tube

As the maximum pressure that can be applied to the muscle and the strength of the muscle depend on the strength of the sleeve and rubber tube, selection of the rubber for the tube and the fiber for the sleeve is very important.

3.1. PBO Fiber Sleeve

p-Phenylene-2, 6-Benzobis Oxazole (PBO) fiber is one of the super fibers that have been developed [12]. The allowable tensile stress of PBO fiber is 5.8 GPa [13]. The strain-stress curve of PBO fiber is shown in **Fig. 3** together with that of three other common fibers: polyester, aramid, and HS-PE (high-strength polyethylene) fibers [13, 14]. The unit of [cN/dtex] is widely used in textile industry, where the unit [cN] is used to represent fiber strength and the unit [dtex] is used to represent the mass of a 10000 m fiber. As shown in **Fig. 3**, PBO fiber is the highest in

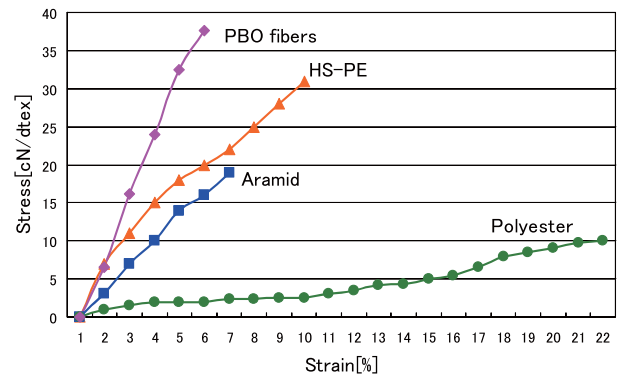


Fig. 3. Strain-stress curve for four types of fiber [12, 13].

Table 1. Characteristic of the chloroprene rubber.

	KR-8	KR-9	KR-10
Hardness (Duro A)	48	60	72
Tensile strength[MPa]	11.6	13.6	12.1
Breaking elongation[%]	430	310	200
Young's modulus[MPa]	0.8	1.3	2.5

strength of the fibers.

The durability of PBO fiber for frictional wear is good compared to that of aramid [13]. Based on these discussions, PBO fiber was selected and applied in this research.

3.2. Chloroprene Rubber

It is necessary to protect the circumference of the sleeve, because A PBO fiber deteriorates with exposure to ultraviolet light and water. In this research, chloroprene rubber applied to new muscle.

Chloroprene rubber stretches like natural rubber, and it resists ultraviolet rays, water, oil, and chemicals. PBO fibers can be protected by covering the sleeve with chloroprene rubber. The characteristics of three types of chloroprene rubber are shown in **Table 1**. KR-8, which has characteristics nearest to those of natural rubber, is applied to the developed artificial muscle.

3.3. Design of Three-Layer Tubes

The three-layer tubes have PBO sleeves inserted between the inner tube and the chloroprene rubber. **Fig. 4** shows a cross section of the actuator. The three-layer tube has an outside diameter of 21 mm, an inner layer of thick rubber 2.0 mm thick, a middle layer of sleeve 1.0 mm thick, and an outer layer of rubber 1.0 mm thick. PBO fibers, which are likely to deteriorate due to water and ultra-violet light, are protected by the inner and outer layers of rubber. Moreover, rubber getting into the gaps in fiber bundles and eliminating clearance between the rubber and sleeve seems to serve to prevent local changes.

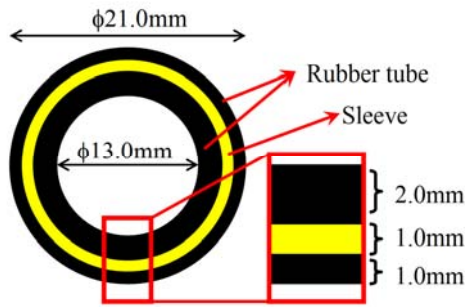


Fig. 4. Cross-section of three-layer rubber tube.

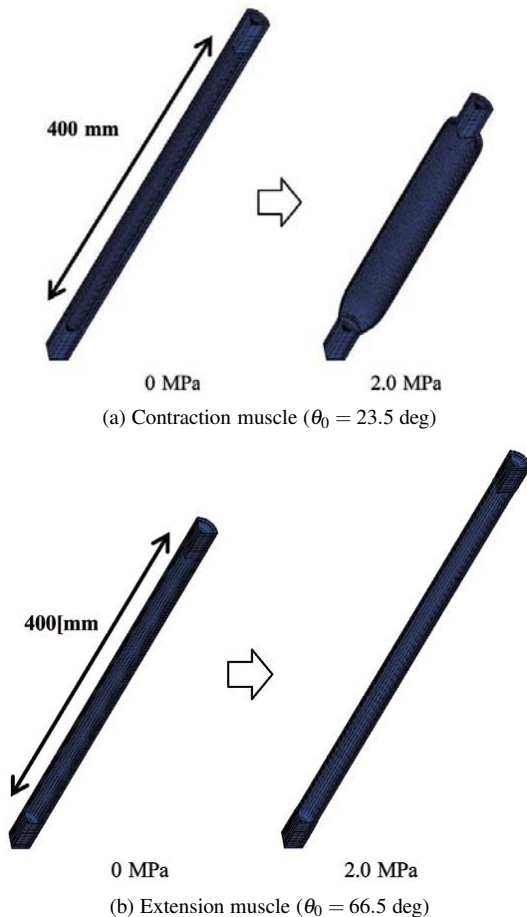


Fig. 5. Nonlinear FEM analysis result.

3.4. Analysis of Three-Layer Tube

Generative forces and variations in artificial muscles depend on the angle of knit of the sleeves [15]. We know from previous studies that artificial muscles that contract maintain a good balance between expansion and contraction when the angle of knit is around 25 deg [16, 17]. We therefore seek the optimum angle of approximately around 25 deg that will provide extension rates identical to contraction rates using the non-linear finite element method.

Figure 5 shows the results of a motion analysis of the two types of the artificial muscles: contraction muscles and extension muscles. Each has the same effective actuation section 400 mm in length, and a pressure of 2.0 MPa

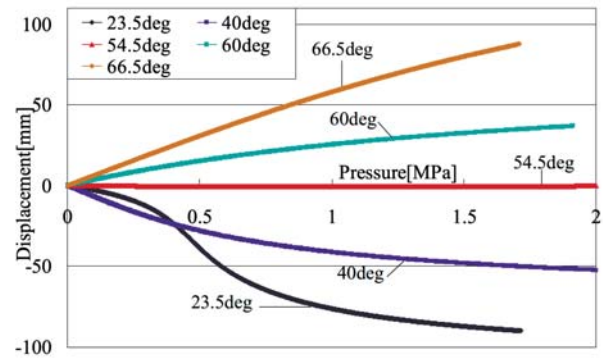
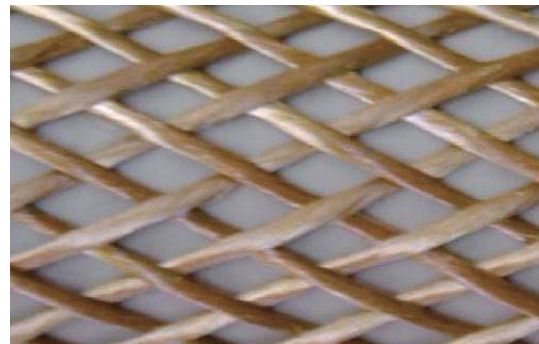


Fig. 6. Relationship between pressure and contraction length of artificial muscle with several angles of knit.



(a) $\theta_0 = 23.5$ deg



(b) $\theta_0 = 66.5$ deg

Fig. 7. PBO fiber sleeve.

is applied. A model of the muscles for analysis by the non-linear, finite element method has an outside diameter of 21 mm and an inside diameter of 13 mm, which are the same dimensions as those of the actual prototype muscles. Figure 6 shows the relation between the applied pressure and the axial directional variations for muscles with several different angles of knit.

Figure 7 shows several muscles with different angle of knit, which can actually be realized by cord-making machines.

Each fiber thickness is 1500 dtex (desi-tex). Contraction muscle sleeve material is made of 48 fiber bundles of PBO fiber with an angle of spiral of 23.5 deg. The thin sleeve bundles are twisted to prevent fluffing. Extension muscle sleeve material is made of 24 fiber bundles of PBO fiber with an angle of spiral of 66.5 deg. Figure 8 shows developed three-layer tube for contraction muscle.



Fig. 8. Developed three-layer rubber tube.

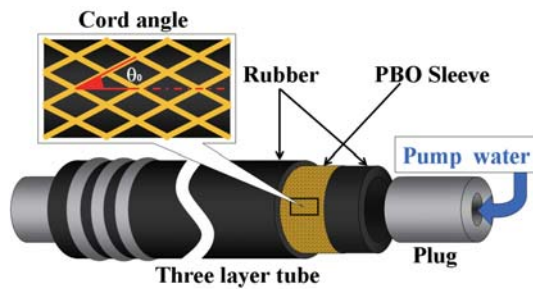


Fig. 9. Structure of developed muscle.

4. Design and Fabrication of Muscle

4.1. Mechanical Structure of Very High-Force Muscle

A special plug to retain the water pressure and the hose band for terminal fixation have been designed and used on the new artificial muscles. The structure of the developed artificial muscles is shown in Fig. 9. Details of the special plug are given in the following paragraph.

4.2. Reversed-Taper Plug

The special plug and the parts of the newly-developed artificial muscle are shown in Fig. 10. The plug has three reversed-taper sections. The three-layer tube on the outside of the plug is bound tight with three hose bands. This reversed-taper structure makes the muscle strong enough to withstand up to 30 MPa hydraulics.

5. E Evaluation Experiments

5.1. Experiment Setup

Figure 11 is an overview of the setup of the experiment to measure generated force. A hydraulic cylinder is used to generate the traction load on the muscle. One end of the muscle is fix; the other end is connected to the cylinder rod. The artificial muscles are driven by a water pump. Water is supplied to the muscle through a hydraulic tube. Two sensors are used, as shown in Fig. 11, to calculate the force generated on the hydraulic cylinder.

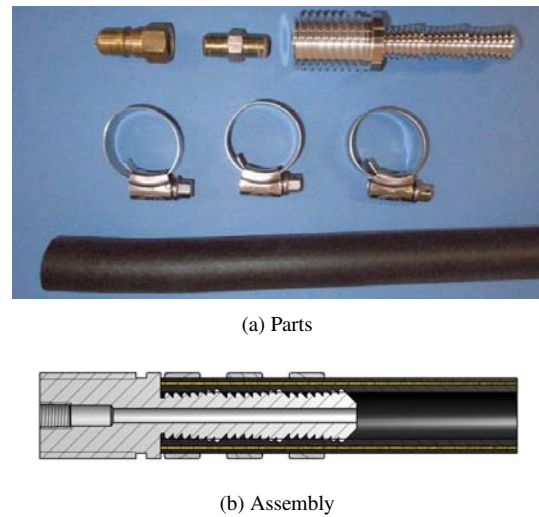


Fig. 10. Assembly of developed muscle.

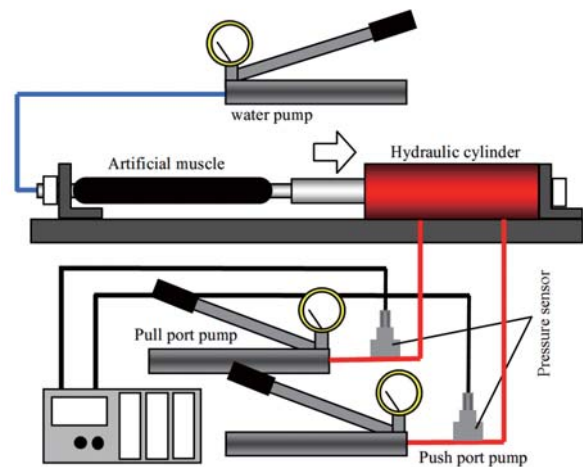


Fig. 11. Setup of force generation test.

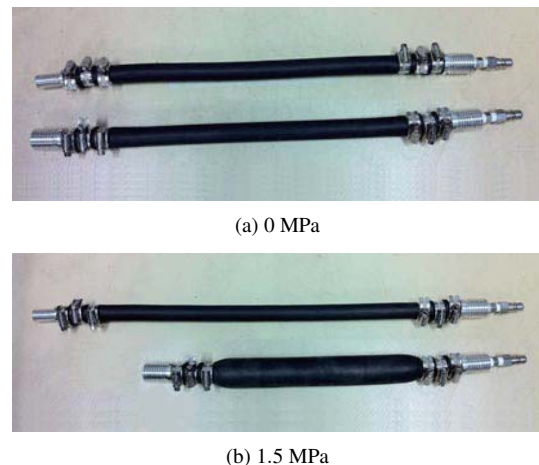
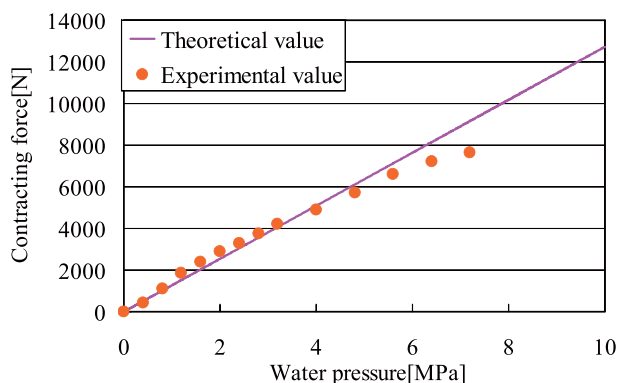
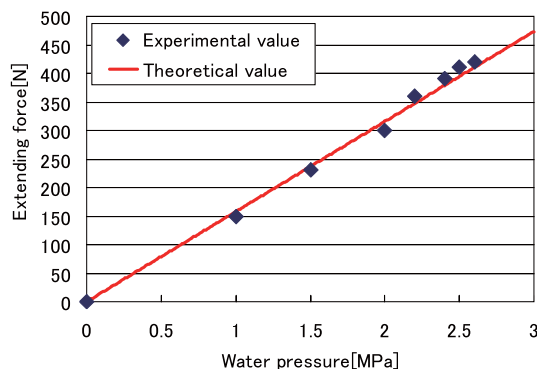


Fig. 12. Driving experiments of thin hydraulic artificial muscles.

In the experiments, we applied 1.5 MPa of hydraulic pressure to each type of artificial muscle, muscles which have effective actuation parts 300 mm in length. Fig. 12 shows the movement of the developed artificial muscles when 1.5 MPa of hydraulic pressure is applied. The arti-

Table 2. Artificial muscle specifications.

	Contraction Muscle	Extension Muscle
Angle[deg] of knit	23.5	66.5
Effective changeable part [mm]	300	300
Displacement[mm] of 1.5 MPa	75	68
Generative force[N] of 1.5 MPa	2285	230
Force per unit cross-section area[N/mm ²]	33	1.8

**Fig. 13.** Comparison of theoretical force of contraction and experimental results.**Fig. 14.** Comparison of theoretical force of extension and experimental results.

ficial muscle on top in the figure is an extension muscle; the one on the bottom is an contraction muscle. Details of the two types of artificial muscles are given in **Table 2** together with their experimental results. Both types of artificial muscle, extension muscles and contraction muscles, realize the variations of around 70 mm in length or nearly equal to the contraction rates obtained from the analysis. Moreover, the contraction muscle can apply a maximum pressure of 7 MPa. The comparisons of theoretical generation forces and experimental results are shown in **Figs. 13** and **14**.

6. Conclusion

In this research, we have developed artificial muscles that contract or extend. In developing such artificial muscles, we have first derived through non-linear FEM analyses the optimum angles for the knit of their sleeves. As a result, we have found that contraction-type artificial muscles with an angle of knit of 23.5 deg and extension-type artificial muscles with an angle of knit of 67.5 deg produce nearly identical variations. We have developed two types of muscles, i.e. contraction muscles and extension muscles, and have conducted experiments to actuate them to confirm that their variations are identical to the analytic results. These results make the prospects of further development of high-power muscles look promising.

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